Exploring the World of Science

2020 Invitational Tournament

WATER QUALITY C [KEY]

This event is sponsored by

STEPHEN M. ROSS SCHOOL OF BUSINESS
UNIVERSITY OF MICHIGAN

Test length: 50 Minutes

Team name:_____

Team number: _____

Student names:

_____

FOR SUPERVISOR: 5pts for Salinometer? _____

PART I (30pts) (Answer Sheet for Powerpoint)

Fill in the common name (1pt) and the region it is found in (0.5pt) for each organism shown:

1. Sea Cucumber, Indo-Pacific
2. Butterfly Fish, Worldwide
3. Crown of Thorns Starfish, Worldwide
4. Banded Coral Shrimp, Worldwide
5. Nassau Grouper, Atlantic
6. Gorgonia, Atlantic
7. Humphead Wrasse, Indo-Pacific
8. Bumphead Parrotfish, Indo-Pacific
9. Flamingo Tongue Snail, Atlantic
10. Parrotfish, Worldwide
11. Barramundi Cod, Indo-Pacific
12. Hard Coral, Worldwide
13. Pencil Urchin, Worldwide
14. Long-spined Black Sea Urchin, Worldwide
15. Snapper, Worldwide
16. Giant Clam, Indo-Pacific
17. Grouper, Worldwide
18. Sponge, Worldwide
19. Fleshy Algae, Worldwide
20. Moray Eel, Worldwide

PART II

1.Using the terms below, fill in the appropriate step/tank in the treatment of wastewater by a wastewater treatment facility (the steps are NOT necessarily in numerical order):

Filtration/Disinfection Primary Clarification

Aeration Basin Screen and Grit Tank Secondary Clarification

A. In the Screen and Grit Tank, solid objects and grit are removed from raw sewage entering the facility
B. In the Aeration Basin, Microbes consume organic matter in an oxygen-rich environment.
C. In Filtration/Disinfection, the water is flushed through charcoal or sand and then treated with either chlorine or UV light
D. Oils, greases, and solids are removed in Primary Clarification.
E. Any remaining oils, greases, and solids are removed in Secondary Clarification.

2. Place the steps from question 1 in chronological order (you may use the letters A-E from question 1).

A D B C E

3. (1pt) Ecological threats to estuaries can include:
a. Coastal Development
b. Water Pollution
c. Overfishing
d. All of the above
e. None of the above

4. (1pt) Choose the *best* description of a benefit of coral reefs:

a. Coral reefs absorb wave energy, providing protection for shorelines
b. Coral reefs absorb wave energy, providing protection for small fish
c. Coral reefs act as a foundation for kelp forest growth
d. Coral reefs draw in carbon dioxide, balancing the ocean's pH levels

5. (1pt) An estuary which is formed when sea levels rise and fall in an existing river valley is said to be a:

a. Coastal Plain Estuary
b. Tectonic Estuary
c. Bar-built Estuary
d. Fjord Estuary

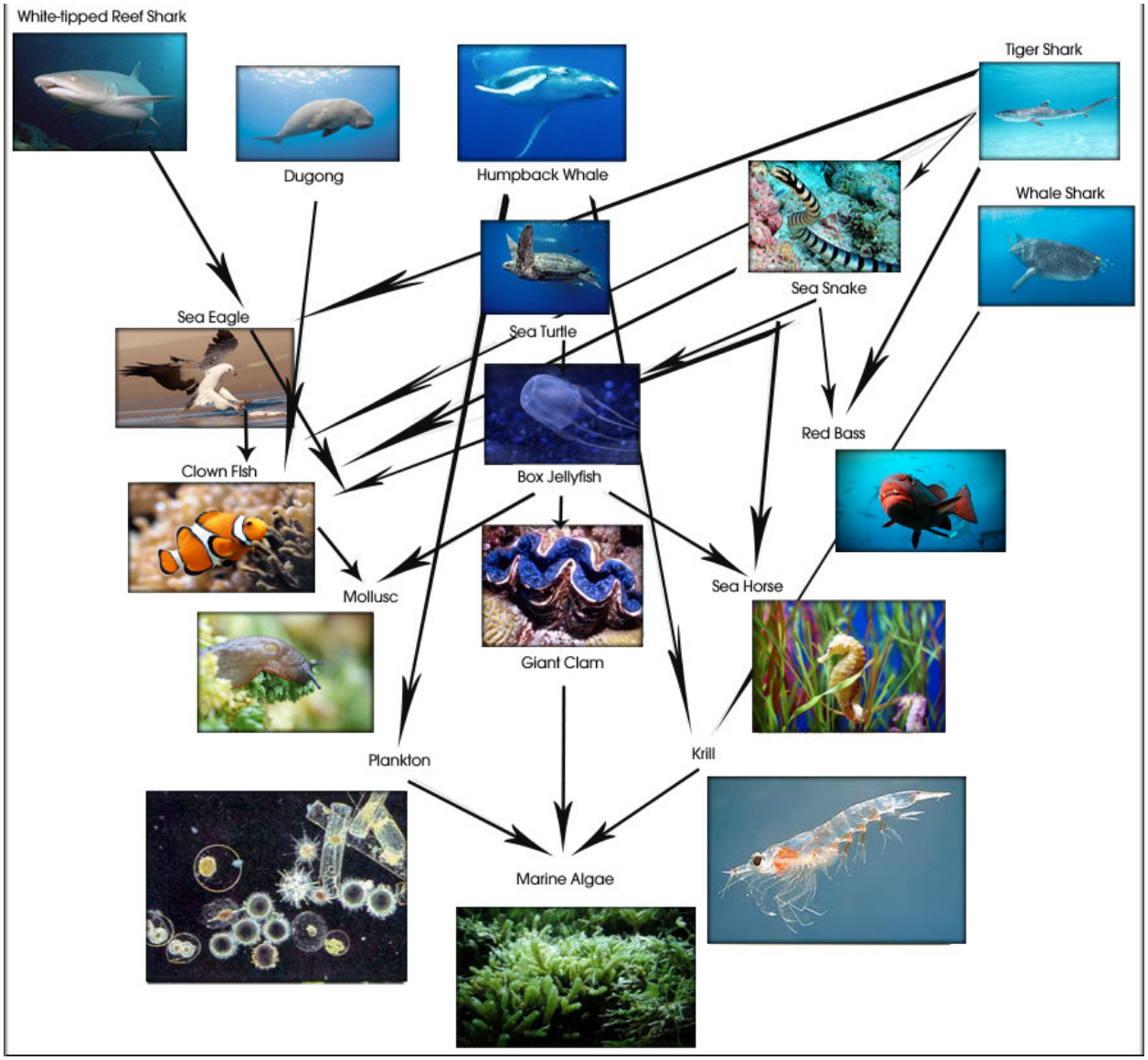
6. (1pt) An estuary which is created when glaciers carve out land and then retreat, allowing seawater to fill the depression is called a:

a. Coastal Plain Estuary
b. Tectonic Estuary
c. Bar-built Estuary
d. Fjord Estuary

7. (1pt) An estuary which is formed by tectonic activity in the Earth's crust is called a:

a. Coastal Plain Estuary
b. Tectonic Estuary
c. Bar-built Estuary
d. Fjord Estuary

The following 3 questions refer to the diagram below



8. The Sea Turtle is considered a **Tertiary** consumer

9. The White-tipped Reef Shark is considered a **Quaternary** consumer

10. (1pt) If the Marine Algae obtains 0.5 Kilojoules of energy from the sun, how many JOULES of energy would the Box Jellyfish obtain from eating the Giant Clam?

a. 50 J
b. 500 J
c. 5000 J
d. 5 J

11. (1pts) Which Acid plays a key role in the degradation of coral reefs?

a. Sulfuric Acid
b. Carbonic Acid
c. Hydrochloric Acid
d. Formic Acid

12. (1pts) Fill in the missing product: (it is formed when carbon dioxide reacts with water and is able to break down coral structures)

$$CO_2 + H_2O \text{ ---> } _H2CO3_$$

13. (1pts) Which of the following statements accurately describe *zooxantheallae*?

a. These organisms live within corals
b. They are a photosynthetic algae
c. *Zooxantheallae* participate in a symbiotic relationship with corals
d. All of the above
e. None of the above

14. (1pts) Which of the following does the NOAA classify as the largest estuary in North America?

a. Chesapeake Bay, Maryland
b. Elkhorn Slough, California
c. Mission-Aransas Estuary, Texas
d. Jobos Bay, Puerto Rico

For the following 8 questions, state whether the statement is true or false (write T for true and F for false in the space provided). If False, replace the underlined term(s) with the correct term..

15. (1pt) TIEBREAKER The spelling of ***zooxantheallae*** (as in question 11) is correct.

F. Zooxanthellae

16. (1pt) Zooplankton are the base of the marine food chain in the ocean's pelagic zone.

F. Phytoplankton

17. (1pt) Coral monochromation is a term used for the color loss of dead/dying corals

F. Bleaching

18. (1pt) Most of the Ocean's primary productivity occurs in the photic zone. T

19. (1pt) El Niño events create rainstorms and flood in typically dry areas and cause T

drought in typically wet areas.

20. (1pt) La Niña events create rainstorms and flood in typically dry areas and cause drought in typically wet areas.

F. El Nino

21. (1pt) The Thermohaline circulation is a global current system including a warm T

surface current and a cold sub-surface current.

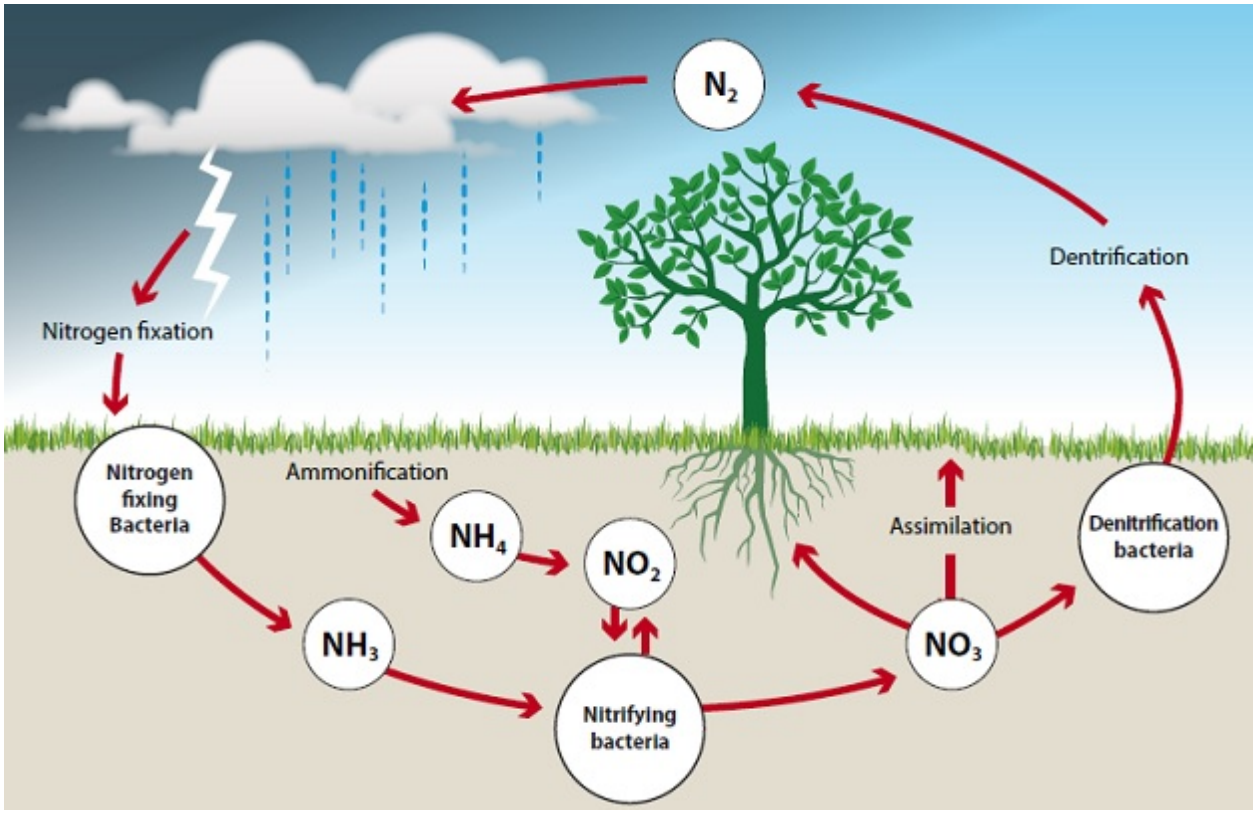
22. (1pt) Downwelled water is rich in nutrients and therefore a site of high productivity.

F, upwelled

23. (4pts) TIEBREAKER: Briefly Explain the difference between water cohesion and water adhesion and give an example of how one of these properties are beneficial to an organism/process.

- Adhesion: the tendency of water molecules to stick to other surfaces (1pt)
- Cohesion: the tendency for water molecules to stick to other water molecules (1pt)
- Surface tension allows insects to skim on top of water, allows for trees to move water from root to leaf, etc. (Reasonable example) (2pts)

24. (2.5pts) Fill in the empty circles with the missing nitrogen-containing molecules which participate in the Nitrogen Cycle



PART III (30pts)

25. (1pt) What is the highest level that Aragonite saturation can reach under normal marine conditions?

a. 2Ω
b. 4Ω
c. 1Ω
d. 0Ω

26. (1pt) A large section of trees which previously covered a body of water is removed in order to expand a local farm by way of placing a cow pen by the shore. *Based only on the tree removal*, which of the following is plausible?

a. Nitrate levels increase
b. Dissolved Oxygen increases from new plant growth
c. Temperature decreases
d. Phosphate levels increase
e. Both (a) and (d)

27. (1pt) The acronym BOD stands for:

a. Biological Oxygen demand
b. Biosolid Oxygen Demand
c. Biochemical Oxygen Demand
d. Biochemical Organism Demand

28. (1pt) Choose the *most* ecologically negative impact of high turbidity as a result of loose sediments

a. Decreased fish spawning
b. Increased aquatic plant growth
c. Unaesthetically cloudy water
d. All of the above

29. (1pt) The average salinity of ocean water is:

a. 4.7%
b. 3.5%
c. 1.6%
d. 5.5%

30. (1pt) High _____ levels can lead to outbreaks of “Blue Baby Syndrome”.

a. Phosphate
b. E-Coli
c. Nitrate
d. Algae

31. (2pts) Calculate the pH of a solution containing roughly 10^9.5 H+ atoms:

Answer: 9.5. 1pt for answer, 1pt for work (logarithmic formula)

pH= -log[H+]

*Vague: + or - 9.5 accepted

32. (3pts) The saturation of Aragonite which is optimal for coral reef growth is higher than the saturation of Aragonite under “normal” marine conditions. Provide an explanation for the ability of water to hold this higher saturation around coral reefs

· Coral reefs are in equatorial regions of the oceans, where water is warmer because of its increased sunlight exposure. Because of this warmer temperature, more Aragonite can be dissolved, making the water supersaturated.

The table below represents the (fictional) results of water quality testing performed on Lake Harbaugh, a small body of water off of the Huron River. Over one summer (July 2016), Farmer Dantonio decides to expand his dairy farm by placing a grazing pasture directly uphill from the lake. One year after Dantonio’s expansion, a restoration project is implemented around the shore of Lake Harbaugh to improve the overall quality of the lake.

	Fecal Coliform (colonies /100mL)	Temperature (Degrees Fahrenheit)	Nitrates (ppm)	Dissolved Oxygen (ppm)	Total Dissolved Solids	Phosphates (ppm)	BOD (mg/ L)
4/1/15	86	58	7	10	60	0.02	3
5/1/15	94	60	6	11	65	0.01	4
6/1/15	102	65	9	15	71	0.03	2
7/1/15	105	61	10	14	62	0.04	5
8/1/15	97	57	7	13	68	0.03	4
4/1/16	82	56	5	12	73	0.01	3
5/1/16	97	59	8	16	69	0.05	2
6/1/16	108	63	6	15	70	0.04	3
7/1/16	236	72	72	3	220	0.25	9
8/1/16	227	74	69	2	214	0.23	10
4/1/17	187	73	67	4	222	0.26	8
5/1/17	215	75	71	2	230	0.22	11
6/1/17	227	76	73	2	215	0.23	12
7/1/17	94	62	11	13	65	0.06	5

8/1/17	81	61	9	15	72	0.02	3

33. (10pts) On the graph paper provided, sketch a complete (label both axes, include a key, and provide a clear title) graph of the nitrate, fecal coliform, and dissolved oxygen measurements based on the data table above.

Needs:

- Accurate pts (-1 for each wrong, otherwise 6pts)
- Title (1pt)
- Labeled axes (1pt each)
- Key (1pt)
- Follows trend (1pt)

34. (3pts) Provide an analysis of the relationship between these three factors based on your graph (Are they directly related? Indirectly? etc.) Explain.

- Nitrate and Fecal coliform are directly related, Dissolved oxygen is inverse to both. 1 pt for each relationship

35.(2pts) What events from the given scenario could explain the relationships you analyzed in the previous question? What could this relationship indicate about the state of Lake Harbaugh?

- Dantonio's farm causes runoff (1pt)
- Lake Harbaugh became Eutrophic (1pt)

36. (2pts) Give a feasible example of a solution that the local restoration team could have implemented in July of 2017.

- Runoff collection/filtration, moving the farm, etc. Must be a reasonable, logical answer for the points

PART IV

37. (10pts) The salinity of the water solution is 3.5%

- 5pts correct salinity
- 5pts for bringing a salinometer